

dSABRE: A SABRE-Style Router for Multi-Core Distributed Quantum Computers

Sanjiang Li *

Abstract—A distributed quantum computer (DQC) executes a circuit across several small processing cores linked by entanglement. Every gate whose operands sit in different cores must first be made local, and each such repair consumes an Einstein–Podolsky–Rosen (EPR) pair—a resource that is orders of magnitude slower to produce than a local gate, and therefore the quantity a distributed router must minimise. Existing SABRE-style distributed routers score the two kinds of repair with unrelated heuristics: a distance heuristic for local SWAPs, and a separate communication-graph energy for inter-core teleports.

We present dSABRE, a SABRE-style router for multi-core processors built on a single principle: both moves are scored as the change they induce in one circuit-wide potential—immediate distance plus depth-discounted lookahead over the circuit’s directed acyclic graph (DAG) of gate dependencies—charged against the resources they spend and penalised for the core capacity they consume. Three consequences follow. Core capacity becomes a term of the score rather than a constraint checked outside it, and ablation shows it contributes more than any other single component. The lookahead set is built in breadth-first DAG layers, which we prove is the ordering that makes a depth discount meaningful and topological filling cannot provide. A checkpoint–rollback escape gives a termination guarantee that a fixed safety-valve cap does not.

Across 21 circuits from the Munich Quantum Toolkit benchmark suite (MQT-Bench) at 25, 36, and 64 logical qubits, dSABRE reduces geometric-mean EPR consumption by 41–46% against TelesABRE, and routes three dense circuits on which it fails to converge. Against the gate-teleportation compiler `pytket-dqc`, dSABRE uses 12–57% fewer EPR pairs on the smaller suites; on the 64-qubit suite `pytket-dqc`’s Steiner-tree embedding is more efficient under unbounded entanglement lifetime, but that advantage inverts once one EPR pair may serve only a bounded number of gates. Results carry over to a non-grid architecture of IBM heavy-hex cores, and a QFT sweep to 360 qubits confirms scalability. Code and online appendices are available at <https://github.com/ebony72/dsabre>.

Index Terms—distributed quantum computing, circuit routing, qubit teleportation, SABRE, EPR minimisation, qubit mapping, lookahead heuristic

I. INTRODUCTION

The near-term path to fault-tolerant quantum computing requires processors with thousands of high-fidelity qubits. No single monolithic chip can economically deliver this scale today. Instead, the field is converging on *distributed quantum computers* (DQC) [?]: multiple small quantum processing units (QPUs, also called cores) interconnected by photonic or microwave links that support remote entanglement generation.

Executing a quantum circuit on a DQC requires a *distributed compilation* pipeline that, like its monolithic counterpart, decomposes into three problem classes [?]. (i) *Qubit allocation* statically assigns logical qubits to cores, typically by hypergraph partitioning [?], [?]. (ii) *Communication-primitive selection* decides how each non-local two-qubit gate is realised—by moving a qubit to its partner’s core, or by executing the gate remotely without moving either [?], [?] (Section II-B describes both primitives and their costs). (iii) *Routing and scheduling* orders the resulting intra-core SWAPs, teleports, and entanglement-generation requests in time, subject to communication-port capacity and finite entanglement rates [?], [?].

All three stages are judged against the same currency. Every communication primitive consumes one or more EPR pairs, and EPR generation is slow (microseconds to milliseconds), noisy, and rate-limited in current hardware [?], [?—orders of magnitude more expensive than a local gate. EPR consumption is therefore the dominant objective of distributed compilation, and the one this paper minimises.

dSABRE addresses the routing component of stage (iii) — ordering intra-core SWAPs and teleports — and leaves scheduling (EPR-generation timing, port reservation) to downstream passes; within stage (ii) it uses only state teleportation, and it is complementary to the static-allocation methods of stage (i). Within this scope the closest prior work is the family of SABRE-style [?] distributed routers, which score teleportation candidates with a heuristic that combines an immediate front-layer distance reduction and a lookahead term over a fixed window of upcoming gates [?]. The most recent and best-performing of these is TelesABRE [?], which uses a Dijkstra computation on a contracted communication graph to estimate inter-core routing cost. We observe one substantive limitation: congestion is handled defensively. TelesABRE discourages landing in saturated cores via a full-core penalty and recovers from deadlock via a safety-valve mode that drops most of the lookahead, but it never proactively *evicts* idle qubits out of crowded cores ahead of demand. On dense or large instances this recovery either burns additional EPR pairs or fails to converge altogether.

Contributions.

- **One scoring principle for both kinds of movement.** dSABRE scores an intra-core SWAP and an inter-core teleport as the same quantity: the change they induce in a single circuit-wide potential—immediate front-layer distance plus depth-discounted lookahead distance—charged against the resources the move spends and penalised for the core capacity it consumes (Section III-A, Propo-

S. Li is with the Centre for Quantum Software and Information, University of Technology Sydney, Australia (e-mail: sanjiang.li@uts.edu.au).

*ORCID: 0000-0002-3332-2546.

sition 1). Prior SABRE-style distributed routers score teleports with a separate communication-graph energy, so their two phases optimise different objectives. Making capacity a term of that one score, rather than a constraint checked outside it, is what carries the result: removing the capacity term inflates geometric-mean EPR by more than any other single change (Section ??).

- **BFS-layer inter-core extended set.** The inter-core scorer builds its lookahead set layer by layer over the remaining DAG so that gates sharing a wire with the front are captured before unrelated wires displace them, with each gate’s BFS depth driving the decay exponent $\gamma^{\text{dep}(g)}$. Proposition ?? shows this ordering is what makes the depth discount meaningful, and that topological filling cannot provide it. Substituting a topological-order extended set increases geometric-mean EPR by 7–11% across the three suites, with the gain growing with circuit size (Section ??).
- **Feasibility mechanisms, reported as such.** Two mechanisms exist to keep the greedy rule deployable rather than to lower EPR count: proactive congestion relief, which teleports idle qubits out of high-demand cores before bottlenecks form, and a checkpoint–rollback escape that forces shortest-path progress on a stuck gate. We report them honestly. Once every core is given reserved escape capacity by the initial layout, relief is neutral on the geometric mean (Section ??); its value, and the escape’s, is completion—dsABRE routes the 64-qubit Random and QAOA circuits and the 200-qubit QFT, on all of which TelesABRE fails to converge. Proposition ?? makes this precise: with the escape in place dsABRE terminates on every circuit, a guarantee a fixed safety-valve cap does not provide.
- **Comprehensive empirical validation.** Across MQT-Bench suites at 25, 36, and 64 qubits on multi-core grid architectures, dsABRE reduces geometric-mean EPR consumption by 41–46% over TelesABRE [?]. It also reduces EPRs by 12–57% over pytket-dqc [?] (strongest Steiner-embedding distributor) on the 25- and 36-qubit suites, while on the 64-qubit suite that method’s gate embedding is more e-bit-efficient on structured circuits—at the cost of ignoring intra-core SWAPs and of runtimes orders of magnitude longer. dsABRE uses a SabreLayout-based initial layout with per-core communication-slot reservation. Ablation studies, cost-ratio sweeps, and a scalability sweep up to 360-qubit QFT confirm that the gains are mainly driven by the heuristic design and congestion mechanisms.

Paper organisation. Section II fixes notation and recaps the SABRE heuristic. Section III presents dsABRE: the routing loop, the intra-core and inter-core scoring rules, the BFS-layer extended set, proactive congestion relief, the checkpoint–rollback escape, and a complexity analysis. Section ?? evaluates dsABRE on the 25q, 36q, and 64q MQT-Bench suites against TelesABRE and pytket-dqc, ablates the design choices, characterises the initial-layout pipeline, reports compile time, and runs a 100–360q QFT scalability sweep. Sec-

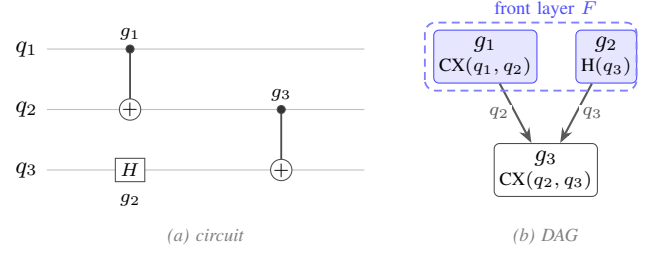


Fig. 1. DAG for a three-gate circuit on qubits q_1, q_2, q_3 . $CX(q_1, q_2)$ and $H(q_3)$ share no qubit and have no incoming edges, so both appear in the initial front layer $F = \{g_1, g_2\}$ and may execute in parallel. $CX(q_2, q_3)$ depends on both via qubits q_2 and q_3 ; it enters F only after g_1 and g_2 have been executed. Circuit depth is 2 (two sequential layers).

tion ?? positions dsABRE against related work; Section ?? outlines directions for further work; Section ?? concludes.

II. BACKGROUND

A. Quantum Circuits

A quantum circuit acts on n qubits through a sequence of gates. *One-qubit (1Q) gates* (Hadamard, rotations, ...) act on a single qubit. *Two-qubit (2Q) gates* entangle pairs: the CX (CNOT) gate is the standard primitive and the building block for most multi-qubit operations. Together, CX and the full single-qubit unitary group $U(2)$ form a universal gate set [?]: every n -qubit unitary decomposes exactly into CX gates and single-qubit rotations. A finite basis such as $\{CX, H, T\}$ suffices for approximate universality and is what the benchmark circuits of Section ?? are transpiled into. The $SWAP$ gate exchanges the states of two adjacent qubits; it decomposes into three CX gates and is used by routing algorithms to migrate logical qubit states along the coupling graph.

A circuit is represented as a *directed acyclic graph* (DAG) in which each node is a gate and each directed edge records a data dependency (qubit produced by gate g_1 consumed by gate g_2). The construction rule is: for every qubit q , connect each gate acting on q to the *next* gate acting on q by a directed edge; the edge means the second gate cannot begin until the first has written its output on q . One-qubit gates therefore contribute at most one outgoing edge per operand, while 2Q gates contribute one per operand pair. Figure 1 shows a small example, where *circuit depth* is the length of the longest path through the DAG.

Physical qubits are arranged in a *coupling graph* $G_{\text{phys}} = (P, E_{\text{phys}})$; a 2Q gate can execute only if its two operands occupy adjacent vertices. A circuit is written in terms of *logical qubits* $q_1, \dots, q_n$; a *layout* $\ell : q_i \mapsto p_p$ assigns each to a physical slot. *Qubit routing* inserts $SWAP$ gates so that the two operands of every 2Q gate reach adjacent physical qubits before it executes.

B. Quantum Teleportation

Quantum teleportation [?] transfers the state of a qubit from a sender to a receiver—in the DQC setting, from one core to another over an inter-core link—using a pre-shared *EPR pair* (a maximally entangled two-qubit state) and two classical bits. The source qubit’s state is destroyed at the sender and

reconstructed at the receiver; no physical qubit travels across the link. Each EPR pair is consumed upon use, so the *EPR count* is the primary inter-core routing cost in distributed quantum compilation. Throughout this paper, “EPR” as a unit (e.g. “10 EPRs”) denotes EPR-pair count; “EPR pair” is used when emphasising the physical entangled state itself.

Two inter-core teleportation variants arise in DQC compilation. *Teledata* moves a logical qubit to a new core (one EPR pair consumed), after which the qubit participates in local gates there. *Telegate* [?] executes a non-local 2Q gate directly across two cores via a shared EPR pair, without physically relocating either qubit. dSABRE uses teledata exclusively; TeleSABRE exploits both variants.

C. Distributed Quantum Architecture

We model a DQC architecture $\mathcal{A}$ abstractly as a weighted graph $G_{\mathcal{A}} = (V, E, w)$ whose vertices are physical qubits. The vertex set is partitioned into K cores $V = V_1 \sqcup \dots \sqcup V_K$; intra-core edges E_{intra} describe in-core connectivity and carry weight 1, and inter-core edges E_{inter} connect designated *communication ports* on neighbouring cores and carry a much larger weight $w_{\text{link}} \gg 1$ reflecting the cost of an EPR pair. A communication port is a physical qubit like any other and may also hold a logical (data) qubit between teleportations; the “communication” label only indicates that the port is one end of an inter-core EPR link, not that the slot is reserved. dSABRE treats $G_{\mathcal{A}}$ as a black box and only queries it through three distance tables: physical d_{phys} , intra-core d_{intra} , and core-graph d_{core} . The router therefore extends to *any* DQC topology — arbitrary intra-core graphs, mixed core sizes, sparse inter-core meshes, hierarchical trees of clusters, etc. — by simply supplying these tables.

For experimental simplicity our reference implementation instantiates $G_{\mathcal{A}}$ as a *grid-of-grids* [?]: each core is an $m \times m$ 2D nearest-neighbour grid, and the cores themselves form an $r \times s$ super-grid joined by a small set of inter-core links between specific boundary qubits. Figure 2 shows two concrete topologies used in our experiments.

a) Hierarchical distance precomputation.: Because intra-core and inter-core costs decouple, the all-pairs distance d_{phys} over $G_{\mathcal{A}}$ admits a two-level Dijkstra computation that is substantially cheaper than running Dijkstra on the full $|V|$ -vertex graph. First, all-pairs distances are computed once *inside* a single core V_i (size $|V_i| = M$) in $O(M^2 \log M)$ time, giving d_{intra} . Second, the *core graph* (a multigraph of K super-nodes connected by inter-core links) is solved in $O(K^2 \log K)$ time, giving d_{core} . Third, $d_{\text{phys}}(p, q)$ for any $p \in V_i, q \in V_j$ is recovered as $d_{\text{intra}}(p, \pi_i^*) + d_{\text{core}}(i, j) + d_{\text{intra}}(\pi_j^*, q)$ minimised over boundary-port pairs (π_i^*, π_j^*) realising the optimal core-graph path. This brings precomputation from $O((KM)^2 \log(KM))$ down to $O(KM^2 \log M + K^2 \log K)$ and is the only architecture-dependent setup dSABRE performs; routing itself uses $O(1)$ table lookups thereafter.

D. Qubit Allocation and Routing

Given circuit DAG C and initial layout $\ell : q_i \mapsto p_{\sigma(i)}$ (σ is a permutation on $0, 1, \dots, n - 1$ with n the number of physical

qubits), routing inserts the minimum number of SWAP and teleportation operations so every gate can execute on adjacent physical qubits. Cost model: each local SWAP costs c_{swap} ; each inter-core teleport costs c_{tele} (one EPR pair).

E. SABRE

LightSABRE [?] optimises SABRE [?] by selecting the SWAP minimising:

$$H = \frac{1}{|F|} \sum_{g \in F} \Delta_F(g) + w \frac{1}{|E|} \sum_{g \in E} \Delta_E(g), \quad (1)$$

where F is the front layer, E the extended lookahead set, and $\Delta(g) = d_{\text{new}}^g - d_{\text{old}}^g$ the change in physical distance between the two qubits of gate g caused by the candidate SWAP (negative when the SWAP moves them closer); d^g is shortest-path distance on the coupling graph. All extended-set gates contribute with equal weight; the $\frac{1}{|E|}$ factor normalises the lookahead term to the same scale as the front term. SABRE also couples its router with an initial-layout optimiser: starting from a random qubit assignment it routes the circuit C forward, then immediately routes the reverse circuit C^{-1} using the final layout of the forward pass as the new starting point; the layout produced by the reverse pass is fed back as the initial layout for the next forward pass. After several bidirectional sweeps the heuristic has annealed towards a layout that is self-consistent with the circuit’s gate structure.

III. dSABRE

dSABRE is a SABRE-style routing algorithm for multi-core processors. Routing assumes a fixed initial layout; we use Qiskit SabreLayout, and defer the discussion of layout preparation and the routing pass schedule to Section ??.

The core invariants and notation are: (i) the layout $\ell : q_q \rightarrow p_p$ assigns logical to physical qubits and the inverse ℓ^{-1} records which logical qubit (if any) currently occupies each physical slot; (ii) the working DAG W holds the remaining gates (initially the full circuit) and shrinks as gates are executed; (iii) the front layer F is the set of gates in W with no unexecuted predecessors, and the extended set E is a fixed-size sliding lookahead beyond F . Table I collects every symbol used in this section.

A. Design Principle: One Potential, Two Scopes

dSABRE moves qubits with two different actions—an intra-core SWAP and an inter-core teleport—but it does not score them with two unrelated heuristics. Both are chosen by the same rule: *take the action that buys the largest reduction in one circuit-wide potential per unit of resource it spends, subject to a penalty for violating core capacity*. This subsection states that potential; the two subsections that follow instantiate it at the two scopes, and every term appearing there is a component of it rather than a separate heuristic.

Fix a layout ℓ and let $d(g; \ell)$ be the weighted distance between the two operands of a 2Q gate g on the full-chip graph $G_{\mathcal{A}}$ (intra-core edges weight 1, inter-core links weight

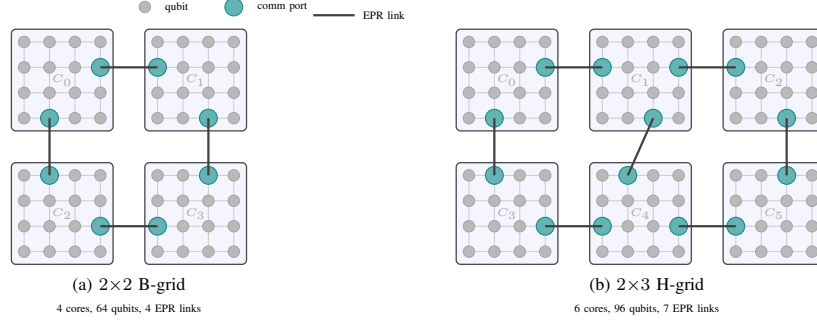


Fig. 2. DQC architectures used in this paper (B-grid and H-grid families introduced by TelesABRE [?]). Small grey circles are qubits; teal circles are inter-core communication ports; bold lines are EPR-channel links between cores. Each core is a 4×4 grid. (a) 2×2 B-grid. (b) 2×3 H-grid.

w_{link}), so $d(g; \ell) = 0$ exactly when g is executable. Define the *routing potential*

$$\Phi(\ell) = \frac{1}{|F|} \sum_{g \in F} d(g; \ell) + w_e \sum_{g \in E} \gamma^{\text{dep}(g)} d(g; \ell), \quad (2)$$

the front-layer term measuring work that must be done now and the lookahead term measuring discounted work that is coming, with the decay $\gamma^{\text{dep}(g)}$ discounting a gate by its depth because a deeper gate is more likely to be displaced by intervening routing decisions before it is reached. Φ is a surrogate, not the true cost-to-go: it is exact at the boundary ($\Phi = 0$ iff every gate in F is executable) and everywhere else it is the standard SABRE relaxation, which ignores that qubits contend for the same slots.

An action a (one SWAP, or one teleport with its staging SWAPs) transforms ℓ into ℓ_a . Write $\Delta\Phi(a) = \Phi(\ell_a) - \Phi(\ell)$, let $\text{cost}(a)$ be the resource the action consumes, and let $\text{pen}(a)$ charge the capacity constraint that Φ omits. dSABRE greedily minimises

$$s(a) = \underbrace{\text{cost}(a)}_{\text{resource spent}} + \underbrace{\text{pen}(a)}_{\text{capacity}} + \underbrace{\Delta\Phi(a)}_{\text{progress bought}}. \quad (3)$$

The three groups answer three different questions—what does this move cost, is it legal-in-practice, and how much closer does it get us—and Eq. 3 is the only place they are traded off. The following proposition, proved by unfolding the definitions, is what makes the rest of this section a specialisation rather than a list.

Proposition 1 (Unified objective): The intra-core SWAP score (Eq. 5) and the inter-core teleport score (Eq. ??) are both instances of Eq. 3. For an intra-core SWAP a inside core c , $\text{cost}(a)$ and $\text{pen}(a)$ vanish and $\Delta\Phi(a)$ restricts to core c ; for a teleport a , $\text{cost}(a) = d_{\text{prep}}$, $\text{pen}(a) = c_{\text{cap}}$, and $\Delta\Phi(a) = \Delta_F + w_e \Delta_E$.

Proof: A SWAP is free in EPR terms and cannot change any core’s occupancy, so $\text{cost} = \text{pen} = 0$ and only $\Delta\Phi$ remains. A SWAP inside c moves no qubit across a core boundary, so $d(g; \cdot)$ is unchanged for every gate with no operand in c ; the surviving terms of Eq. 2 are those over F_c and E_c , and dividing by the constants $|F_c|, |E_c|$ (a positive rescaling, which does not change the argmin) gives exactly Eq. 5. A teleport consumes one EPR pair after d_{prep} staging SWAPs, which is its resource term, and changes the occupancy

of the landing core, which is what c_{cap} charges; it relocates the single qubit q_1 , and $\Delta_F + w_e \Delta_E$ is by definition the induced change in the two sums of Eq. 2, giving Eq. ??.

Two consequences are worth stating, because they are what the experiments later test. First, the capacity term is *not* a tie-breaker bolted onto a distance heuristic: it is the one place where the architecture’s finiteness enters a score that is otherwise a pure relaxation, and Section ?? finds it the single largest contributor to dSABRE’s advantage. Second, because the same Φ is used at both scopes, a lookahead gate influences an intra-core SWAP and an inter-core teleport through the same discounted distance, so the two phases cannot pull against each other—which is what distinguishes dSABRE from routers that score teleports with a separate communication-graph energy (Section ??).

The mechanisms of Sections ?? and ?? are deliberately *not* additional score terms. They exist because a greedy minimiser of Eq. 3 can reach states where Φ stops being a usable surrogate—every cheap action blocked by saturated cores, or no action reducing $|W|$ at all. Congestion relief and checkpoint-rollback keep the router inside the region where Eq. 3 means what it says, and we report their value in those terms (feasibility and completion) rather than as EPR reductions.

With the principle fixed, at the intra-core level dSABRE’s SWAP-selection objective inherits the front-layer / extended-set scoring of Eq. 1, with two refinements: a decay-weighted lookahead ($\gamma^{\text{dep}(g)}$ down-weights deep successors) and taint propagation that excludes qubits already entangled with cross-core gates (Section III-C). Inter-core movement is handled by a separate teleportation scorer (Section III-D).

B. Workflow

Algorithm 1 and Figure ?? sketch the main routing loop. Each iteration runs four phases: **(P1)** drain F (execute 1Q gates and any 2Q gate whose operands are already adjacent in the same core, iterated to a fixed point); **(P2)** partition the remaining front into F_{intra} (both operands in one core) and F_{inter} (operands in different cores); **(P3)** if $F_{\text{intra}} \neq \emptyset$ apply one intra-core SWAP (Section III-C), else score teleportation candidates over F_{inter} together with proactive relief candidates and apply the cheapest (Section III-D); **(P4)** checkpoint (ℓ, W) if $|W|$ decreased, otherwise trigger checkpoint-rollback after

TABLE I
NOTATION USED THROUGHOUT SECTION III.

Symbol	Meaning
ℓ, ℓ^{-1}	layout logical→physical, and its inverse
W	working DAG of unexecuted gates
F, E	front layer; extended (lookahead) set
F_c, E_c	their restrictions to core c
$G_{\mathcal{A}}$	full-chip graph; inter-core links weight w_{link}
$d(g; \ell)$	weighted distance between the operands of gate g
$d_{\text{intra}}, d_{\text{core}}$	intra-core and core-graph distance
$\text{dep}(g)$	depth of g (BFS layer for the inter-core set)
Φ	routing potential, Eq. 2
Δ_F	change in d over the front gate, Eq. ??
Δ_E	decay-weighted change over E , Eq. 4
d_{prep}	staging + eviction SWAPs before a teleport
c_{cap}	capacity penalty; τ threshold, c_{pen} multiplier
f_{dst}	free slots in the landing core c_{next}
π_s, π_d	source and destination comm ports of a link
n_s	staging slot: neighbour of π_s closest to p_1
w_e, γ	lookahead weight; lookahead decay
K, M	number of cores; qubits per core

L_{deadlock} stalled iterations and abort after $N_{\text{backup}}^{\text{max}}$ failed recoveries. The two scoring functions operate at different scopes: the intra-core scorer is per-core, restricted to intra-core continuations (qubits involved in inter-core gates are tainted), and front-/extended-set averaged in the SABRE style; the inter-core scorer evaluates moves across cores using the global F_{inter} and a global BFS-layer extended set (Section ??), with Δ_E entering as an un-averaged decay-weighted sum because the cross-core lookahead window is small and its size already enters implicitly via the $|E|_{\text{max}}$ cap.

C. Intra-Core SWAP Heuristic

When the front contains a gate whose operands share a core but are not adjacent, dSABRE chooses an intra-core SWAP using the standard SABRE heuristic, restricted to the relevant core. Let $F_c = \{g \in F_{\text{intra}} : \text{core}(g) = c\}$ be the front gates in core c and $E_c = (g_0, g_1, \dots)$ the corresponding intra-core extended set, an *ordered list* of at most L gates (default $L=20$) collected in topological order. The traversal stops including a gate the moment either of its qubits has previously appeared in a cross-core gate in the working DAG (taint propagation), ensuring E_c contains only gates that remain local to core c . Each gate g_i is assigned a depth $\text{dep}(g_i)$, the number of intra-core gates on the longest qubit-wire path from the front layer to g_i (front layer = depth 0; first successor = depth 1). The lookahead contribution is

$$\tilde{\Delta}_E^c = \sum_{i=0}^{|E_c|-1} \gamma^{\text{dep}(g_i)} (d_{\text{new}}^{g_i} - d_{\text{old}}^{g_i}), \quad (4)$$

where $\gamma \in (0, 1]$ is the lookahead decay and the distance terms use the intra-core graph. The tilde marks the departure from SABRE's flat extended-set weighting: rather than treating all lookahead gates equally, the exponential factor $\gamma^{\text{dep}(g_i)}$ down-weights gates far from the front so that near-term successors dominate the lookahead signal — a deeper gate is more

Algorithm 1 dSABRE route(C, ℓ_0)

```

1:  $W \leftarrow$  working copy of circuit DAG  $C$ ;  $\ell \leftarrow \ell_0$ 
2:  $C_{\text{out}} \leftarrow$  empty routed circuit on the physical qubits
3: save checkpoint ( $\ell, W$ )
4: while  $W$  has unexecuted gates do
5:   drain  $F$  (execute 1Q gates and adjacent intra-core 2Q
     gates; append them to  $C_{\text{out}}$ ) (P1)
6:   if  $W$  empty then
7:     break
8:   end if
9:   compute  $F$ , partition into  $F_{\text{intra}}, F_{\text{inter}}$  (P2)
10:  if  $F_{\text{intra}} \neq \emptyset$  then
11:    compute per-core intra-only extended set  $E_c$  for each
      active core  $c$  (P3a)
12:    pick best SWAP  $(u, v)$  within a core (Eq. 5)
13:    apply SWAP( $u, v$ ) to  $\ell$ ; append SWAP( $u, v$ ) to  $C_{\text{out}}$ 
14:  else if  $F_{\text{inter}} \neq \emptyset$  then
15:    compute global  $E$  if not cached (P3b)
16:    generate candidates over  $F_{\text{inter}}$  and relief moves
17:    pick lowest-score candidate (Eq. ??)
18:    apply teleport to  $\ell$ ; append teleport (staging SWAPs
      + teledata) to  $C_{\text{out}}$ 
19:  end if
20:  if  $|W|$  decreased then
21:    save checkpoint (P4)
22:  else if  $L_{\text{deadlock}}=50$  stalled iterations then
23:    restore checkpoint; backup_plan( $W, \ell, C_{\text{out}}$ ) //
      abort after  $N_{\text{backup}}^{\text{max}}=50$ 
24:  end if
25: end while
26: return ( $\ell, C_{\text{out}}, \text{metrics}$ )

```

likely to be displaced by intervening routing decisions and so contributes less reliable information about the right SWAP now. The same weighted scheme is reused by the inter-core scorer (Section III-D); Section ?? explains how $\gamma^{\text{dep}(g)}$ interacts with the BFS-layer extended set, where $\text{dep}(g)$ is the BFS depth rather than an iteration index. For every neighbour pair (u, v) within a core, the score

$$H_{\text{intra}} = \frac{\Delta_F^c}{|F_c|} + w_e \frac{\tilde{\Delta}_E^c}{|E_c|} \quad (5)$$

is the SABRE objective for intra-core scoring with the lookahead-decay weighting of Eq. 4. The lowest-score SWAP is applied; tie-breaking is by enumeration order. Because the scorer is restricted to a single core and the candidate set is the $O(M)$ intra-core neighbour pairs, the per-core selection is trivially parallel across active cores.

D. Inter-Core Teleportation Scoring

For each gate $g \in F_{\text{inter}}$ on logical qubits q_1, q_2 , let c_{src} and c_{tgt} be the cores currently hosting q_1 and q_2 respectively (i.e. $\ell(q_1) \in c_{\text{src}}$ and $\ell(q_2) \in c_{\text{tgt}}$, where ℓ is the current layout). Both endpoints are tried as teleportation sources. For the qubit chosen as source (say q_1 at physical position $p_1 = \ell(q_1)$), the router enumerates every neighbouring core c_{next} and every

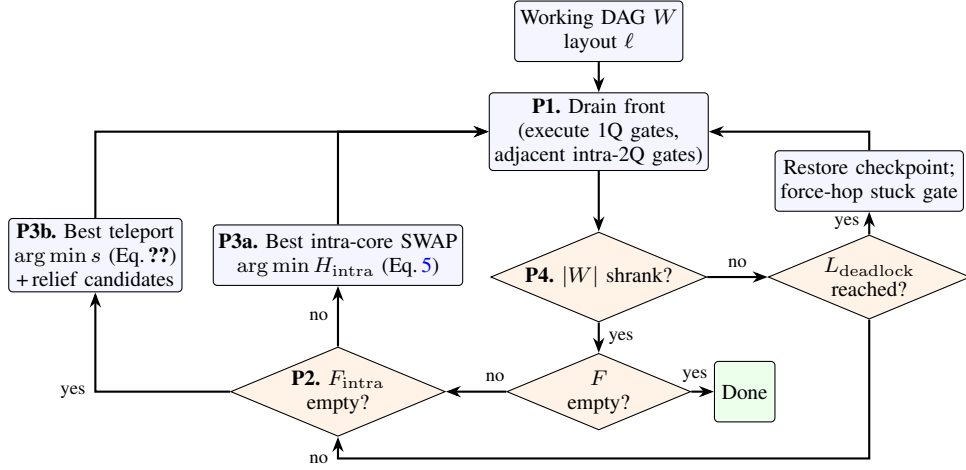


Fig. 3. The dSABRE routing workflow. Each iteration drains the front layer (P1), classifies the remaining front into intra-core and inter-core gates (P2), applies one SWAP or one teleport (P3), and checkpoints when forward progress is made (P4).

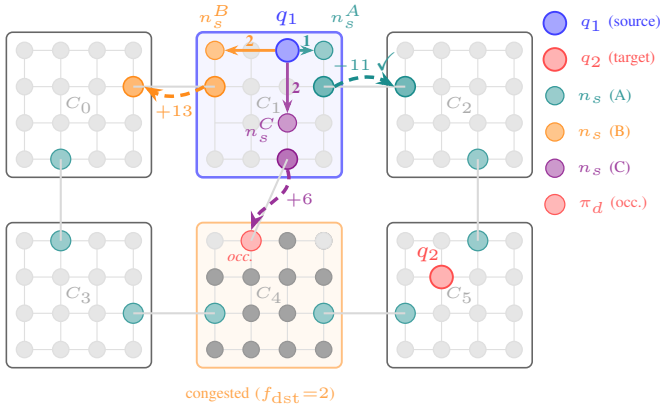


Fig. 4. How dSABRE chooses where to teleport a qubit. A single cross-core gate $g=(q_1, q_2)$ is pending, with q_1 in core C_1 (blue border) and its partner q_2 in C_5 ; the router scores the three links leaving C_1 and takes the cheapest by Eq. ???. Executing any one of them takes two steps: intra-core SWAPs first move q_1 to the staging slot n_s (count on the solid arrow), then a teleport along the link (π_s, π_d) carries it into the next core (dashed arc, annotated with the resulting score s). The three candidates isolate one decision factor each, and correspond row by row to Table ???. **A** (teal) moves towards C_5 and wins, $s_A = -11$. **B** (orange) moves to C_0 , away from C_5 , so its front-layer term turns positive. **C** (violet) makes the same progress as **A** but lands in the nearly-full core C_4 (orange border, two free slots), where the capacity penalty outweighs the progress. The red port marked *occ.* is already occupied, so **C** also pays an extra eviction SWAP.

inter-core link (π_s, π_d) between c_{src} and c_{next} . Each such triple $(q_1, c_{next}, (\pi_s, \pi_d))$ is one *candidate*; Figure ?? illustrates the structure. Executing a candidate involves two steps: (1) intra-core SWAPs that move q_1 from p_1 to the *staging slot* n_s , the neighbour of π_s closest to p_1 ; and (2) a teleportation that consumes the EPR pair on link (π_s, π_d) and delivers q_1 from n_s to π_d in c_{next} . Each candidate is scored as:

$$s = \underbrace{d_{\text{prep}}}_{\text{staging}} + \underbrace{c_{\text{cap}}}_{\text{capacity}} + \underbrace{\Delta_F + w_e \tilde{\Delta}_E}_{\Delta\Phi \text{ (front + lookahead)}}, \quad (6)$$

where lower scores are preferred. The three groups are the three groups of Eq. 3: what the move spends, whether it is

legal-in-practice, and what it buys. Each term is defined as follows.

Staging cost d_{prep} is the number of intra-core SWAPs needed to move q_1 from p_1 to staging slot n_s , plus eviction SWAPs to free the comm ports if occupied:

$$d_{\text{prep}} = d_{\text{intra}}(p_1, n_s) + d_{\text{evict}}(\pi_s) + d_{\text{evict}}(\pi_d).$$

Capacity penalty $c_{\text{cap}} = c_{\text{pen}} \max(0, \tau - f_{\text{dst}})$ discourages routing into nearly-full cores. f_{dst} is the number of free slots in the immediate landing core c_{next} (not the final partner-qubit core c_{tgt}), τ the capacity threshold, and c_{pen} a cost multiplier. This prevents progressive crowding of hot cores that leads to deadlock.

Front-layer change $\Delta_F = d_{\text{new}}^g - d_{\text{old}}^g$ is the change in weighted physical distance between q_1 and q_2 after the teleport (negative when they are brought closer), computed on the full-chip graph G_A (intra-core edges weight 1, inter-core links weight w_{link}). There is exactly one front-layer gate involving the moving qubit q_1 : because any two gates that share a qubit are ordered by a dependency edge in the DAG, at most one of them can be unblocked (i.e. in F) at any time, and $g = (q_1, q_2) \in F_{\text{inter}}$ is already that gate by the premise of this scoring step.

A term we removed. The conference version of this scorer carried a fifth, position-independent directional reward $g_{\text{hop}} = w_h(d_{\text{core}}(c_{\text{src}}, c_{\text{tgt}}) - d_{\text{core}}(c_{\text{next}}, c_{\text{tgt}}))$, intended to supplement Δ_F when the landing port π_d sits far from the best staging position for the next hop. Proposition 1 explains why it is redundant: Δ_F is already the exact change in Φ , and Φ measures distance on G_A , whose inter-core links carry weight $w_{\text{link}} \gg 1$ —so a move to a core one hop closer to the target already registers a gain of order w_{link} , and g_{hop} re-counts it. Ablation confirms the redundancy empirically: removing g_{hop} changes the 25- and 36-qubit geometric means by 0.0%—every circuit ties exactly—and the 64-qubit suite by less than seed noise (Section ??). We therefore drop the term, which removes one hyperparameter and leaves Eq. ?? in exact correspondence with Eq. 3. All numbers reported in this paper are for the four-term score.

Decay-weighted lookahead. $\tilde{\Delta}_E$ is the same decay-weighted lookahead introduced in Section III-C (Eq. 4), reused here with two adaptations for the inter-core scorer: E is now the inter-core extended set (Section ??) rather than the per-core taint-propagated list, and the sum is restricted to gates involving the teleported qubit, since only those gates are affected by the candidate move. The same γ controls how rapidly deeper successors are discounted.

a) *Running example.*: Figure ?? illustrates Eq. ?? on the 2×3 H-grid with defaults $w_{\text{link}}=10$, $\tau=3$, $c_{\text{pen}}=15$. Gate $g=(q_1, q_2)$ has $d_{\text{core}}(C_1, C_5)=2$ and is the only pending gate on q_1 , so Δ_F is evaluated from g alone and $\tilde{\Delta}_E=0$ (empty extended set). With $p_1=(0, 2) \in C_1$ and $p_2=(1, 1) \in C_5$, the shortest path on $G_{\mathcal{A}}$ gives $d_{\text{old}}^g=28$ (via $C_1 \rightarrow C_2 \rightarrow C_5$). Table ?? scores the three outgoing-link candidates, and shows the three groups of Eq. 3 deciding between them: A wins by combining the cheapest staging with the largest progress ($\Delta_F=-12$); B loses because q_1 moves away from C_5 and Δ_F turns positive; C buys the same progress as A but lands in a nearly-full core, and the capacity term alone reverses the decision.

TABLE II
SCORE BREAKDOWN FOR THE THREE CANDIDATES OF THE RUNNING EXAMPLE (FIGURE ??). LOWER IS BETTER; CANDIDATE A IS SELECTED.

Candidate	next core	direction	d_{prep}	c_{cap}	Δ_F	score s
A	C_2	towards C_5	1	0	-12	-11
B	C_0	away from C_5	2	0	+11	13
C	C_4	towards C_5	3	15	-12	6

E. Inter-Core Extended-Set Construction

The lookahead $\tilde{\Delta}_E$ in Eq. ?? depends on how the inter-core extended set E is built. SABRE [?] fills E in topological order, interleaving all wires uniformly; TelesABRE [?] uses BFS layers but within each layer emits gates in arbitrary order and weights each by its iteration index g via $(1 + g/10)$ rather than by depth. Both schemes can deprioritise the very follow-on gates a candidate teleport would help, causing $\tilde{\Delta}_E$ to undercount its benefit.

dsABRE keeps a BFS-layer expansion of the remaining DAG with two refinements: (i) within each layer, gates sharing a qubit with the front are emitted first; (ii) every gate in BFS layer k is assigned $\text{dep}(g)=k$, used in the decay $\gamma^{\text{dep}(g)}$, so the size- L truncation cuts cleanly at a layer boundary and weighting decays exponentially with depth rather than growing linearly with iteration index. It adds no hyperparameter and runs in $O(|E|_{\text{max}})$ time per call when remaining in-degrees are maintained incrementally: BFS peels zero-in-degree gates layer by layer, each emitted gate triggering $O(1)$ decrements on its successors. The empirical contribution over the topological variant is reported in Section ??.

The reason the construction matters is structural rather than incidental. In Eq. 2 a gate's influence is set by $\gamma^{\text{dep}(g)}$, so the lookahead is only meaningful if the gates that survive the size- L truncation are the shallow ones. BFS layering guarantees exactly that; topological order does not.

Proposition 2 (Depth monotonicity): Let E_{BFS} and E_{topo} be the size- L extended sets built by BFS-layer expansion and by topological order. Then E_{BFS} contains every gate of depth $\leq k$ before any gate of depth $> k$, for every k . No topological filling has this property: for every L there is a DAG in which E_{topo} omits a depth-1 gate while admitting L gates of depth up to L .

Proof: The first claim is the defining property of breadth-first expansion: layer $k+1$ is emitted only after layer k is exhausted. For the second, take $L+1$ disjoint qubit wires; on wire 0 place a chain of L 2Q gates $g_1, \dots, g_L$ with $\text{dep}(g_i)=i$, and on wire 1 a single 2Q gate h with $\text{dep}(h)=1$ whose predecessor is the front gate of that wire. A topological order may emit the entire chain of wire 0 before visiting wire 1, filling the size- L budget with $g_1, \dots, g_L$ and excluding h ; BFS emits g_1 and h in layer 1. The excluded h is a gate that becomes executable one layer ahead, so its distance term is the one a teleport decision most needs. ■

This is why the gain from BFS layering grows with circuit size (Section ??): the deeper and wider the remaining DAG, the further a topological sweep drifts from the active front before the budget L is spent. The proposition bounds the ordering, not the EPR count—routing quality still depends on the circuit—but it identifies the mechanism the ablation measures.

F. Proactive Congestion Relief

In dense layouts, a core can become saturated: many pending gates demand qubits from it simultaneously and the core has no free slots to absorb incoming teleports. Standard reactive scoring (Eq. ??) does not address this until the bottleneck triggers a deadlock.

dsABRE adds a *proactive congestion relief* mechanism that fires alongside the gate-driven scoring of Eq. ??.

Demand vector. At each iteration the router computes a demand vector $d[c]$ over all cores by counting inter-core gates in the front and lookahead windows whose shortest core-graph path traverses c . $d[c]$ is therefore an upper bound on the number of upcoming teleports that will need to pass through c , regardless of whether c itself hosts an operand.

Trigger. A core c is flagged when both $d[c] \geq \theta_d$ (default $\theta_d=3$) and $f_c \leq \theta_f$ (default $\theta_f=2$) hold: high incoming demand combined with low free capacity. The reactive scorer alone would only respond once f_c hits zero and the next gate-driven teleport fails to find a landing port; firing earlier on the conjunction avoids the eviction-cascade pattern that triggers deadlock.

Relief candidates. For each flagged core, the router generates teleports that move the most-idle logical qubits in c — those whose next pending gate has the largest DAG depth from the front — into a less-loaded neighbour. Each candidate is scored by Eq. ?? with $\Delta_F=0$ — victims are restricted to qubits absent from the front layer, so no pending gate motivates the move. The decay-weighted lookahead $\tilde{\Delta}_E$ is retained, so that when a victim does have an upcoming gate within the extended-set window the landing port is preferred to be closer to its next partner. A *relief bonus* $-b_r \cdot (d[c] - f_c)$

proportional to the demand–capacity imbalance is added on top. Relief candidates compete directly against gate-driven candidates and win only when the bonus outweighs the staging cost; the mechanism therefore introduces no extra teleports when the front-layer scorer already has a cheap on-target move available. On the per-core adaptive layout the ablation in Section ?? shows relief is close to gmean-neutral (disabling it *lowers* 64q gmean EPR by 8.7%): the layout now gives every core escape capacity, so the standing-congestion the relief mechanism was designed to pre-empt is largely absent, and its remaining contribution is confined to individual hard instances such as the 64q Random circuit.

G. Checkpoint–Rollback Deadlock Escape

When Eq. ?? stalls, dSABRE falls back to LightSABRE’s release-valve idea [?]: force shortest-path progress on the most-stuck front-layer gate. We wrap it in a checkpoint: every iteration that reduces $|W|$ saves (ℓ, W) , and after L_{deadlock} stalled iterations the router restores the checkpoint before forcing intra-core SWAPs or hop-by-hop teleportation along $\text{core_path}(c_1, c_2)$, aborting after $N_{\text{backup}}^{\text{max}}$ failed recoveries. The escape is dormant on best-EPR runs and only salvages individual SabreLayout seeds on dense large circuits (e.g. one of three seeds on 200q QFT, with 17 activations).

Its purpose is not to lower EPR count but to make the greedy rule of Eq. 3 safe to deploy. A pure greedy minimiser has no termination guarantee: nothing in Eq. 3 prevents a cycle of moves that each look locally attractive while $|W|$ never shrinks. The forced-progress escape removes that possibility.

Proposition 3 (Progress and termination): Let D be the diameter of the core graph and suppose the rollback budget $N_{\text{backup}}^{\text{max}}$ is not exhausted. Then dSABRE executes every gate of the circuit, and the forced escape spends at most D EPR pairs on the gate it unblocks.

Proof: Consider any state in which no gate has been executed for L_{deadlock} iterations, and let $g = (q_1, q_2)$ be the stuck front gate the escape selects. The escape ignores the score and teleports q_1 hop by hop along $\text{core_path}(c_1, c_2)$, a shortest path in the core graph, evicting an occupant at each landing core if necessary. Each hop strictly decreases $d_{\text{core}}(\text{core}(q_1), c_2)$, so after at most $d_{\text{core}}(c_1, c_2) \leq D$ hops—hence at most D EPR pairs—the two operands share a core, and the intra-core SWAP heuristic then makes g executable in at most d_{intra} -many further SWAPs, bounded by the core diameter. So $|W|$ strictly decreases within a bounded number of iterations of any stall, and since $|W|$ is a non-negative integer that never increases, the loop terminates with $W = \emptyset$. Eviction cannot block the argument: the reserved escape slots of Section ?? guarantee a free slot in the landing core, and where none remains the evicted qubit is displaced within its core rather than across a boundary, leaving d_{core} unchanged for q_1 . ■

The deployed configuration caps recoveries at $N_{\text{backup}}^{\text{max}} = 50$, so dSABRE inherits the guarantee up to that budget and reports an abort beyond it; no run in this paper reached the cap. TelesABRE offers a fixed safety-valve iteration cap instead of a progress argument, which is consistent with its failures to converge on the denser circuits of Section ??.

H. Computational Complexity

Let $N = |C|$, n be the circuit width, K the number of cores, M qubits per core ($P = KM$), and L the extended-set capacity (constant). Architecture distance tables are precomputed once in $O(KM^2 \log M + K^2 \log K)$ (Section II-C) and are not charged per circuit.

Per iteration, inter-core scoring generates $O(n)$ candidates each costing $O(L)$, and intra-core scoring evaluates $O(M)$ SWAP candidates per core at cost $O(|F_c| + L)$, summing to $T_{\text{iter}} = O(Mn + P)$. In the common case $n \geq K$ (e.g. $n/K \in \{6.25, 9.0, 10.7\}$ on our suites) this is $O(Mn)$. The extended set is rebuilt at most N times at $O(KL)$ each, adding $O(NK)$ total. On a grid-of-grids the physical diameter is $O(\sqrt{P})$, bounding inserted operations by $O(N\sqrt{P})$; the checkpoint–rollback escape adds a constant $N_{\text{backup}}^{\text{max}} \cdot L_{\text{deadlock}}$ iterations. Combining,

$$T_{\text{total}} = O(N\sqrt{P} \cdot Mn). \quad (7)$$

Compared to flat SABRE [?] on the same P qubits ($O(Pn)$ per iteration), restricting intra-core scoring to core-local edges yields a factor- K per-iteration improvement that grows with the number of cores. Empirically this scales well in practice: the largest circuit we route (360-qubit QFT with 13,300 CX gates on a 486-qubit H-grid) routes in 801 s of pure-Python wall time (Section ??).

IV. EXPERIMENTAL EVALUATION

A. Setup

Benchmarks. We use three circuit sets from the MQT Bench suite [?], all in the IBM native-gate set (Qiskit opt3 transpilation, measurements and barriers removed).

Suite 1 (25-qubit): quantum amplitude estimation (AE), quantum Fourier transform (QFT), quantum neural network (QNN), random circuit (Random), GHZ state preparation, and graph-state preparation.

Suite 2 (36-qubit): six circuits spanning a broader range of algorithmic families and CX densities: Bernstein–Vazirani (BV, linear chain), Deutsch–Jozsa (DJ, oracle), W-state preparation (cascaded star), VQE with SU(2) ansatz (layered variational), QAOA (dense all-to-all), and quantum phase estimation (QPEexact, hierarchical).

Suite 3 (64-qubit): the same six circuit families as Suite 1 scaled to 64 logical qubits (AE, QFT, QNN, Random, GHZ, Graphstate), run on the larger H-grid architecture, *extended* with three further circuits that complete the algorithmic-class coverage at the largest scale: QPEexact (phase-estimation family, the class on which gate teleportation is strongest), QAOA (dense max-cut cost layers with shared-control ZZ structure), and a 16-bit Multiplier (arithmetic, the suite’s largest circuit at 13 040 CX). QPEexact is the MQT-Bench v1.1.0 nativegates/opt3 circuit; QAOA and Multiplier are generated with MQT-Bench v2.2.2 in the same format (basis $\{R_z, SX, X, CX\}$, Qiskit optimisation level 3).

Circuit properties are summarised in Table ??.

Architecture. Suites 1 and 2 use a 2×2 B-grid: four 4×4 cores (64 physical qubits, 4 EPR links). Suite 3 uses a 2×3

TABLE III
BENCHMARK CIRCUIT PROPERTIES (MEASUREMENTS/BARRIERS
STRIPPED)

Suite	Circuit	Qubits	CX gates	Depth	CX/qubit
25q	AE	25	558	395	22.3
	QFT	25	580	173	23.2
	QNN	25	1223	259	48.9
	Random	25	1124	589	45.0
	GHZ	25	24	27	1.0
	Graphstate	25	25	19	1.0
36q	BV	36	17	23	0.5
	DJ	36	35	41	1.0
	W-state	36	70	145	1.9
	VQE-SU2	36	105	56	2.9
	QPEexact	36	1019	347	28.3
	QAOA	36	1200	256	33.3
64q	AE	64	1962	1058	30.7
	QFT	64	1966	446	30.7
	QNN	64	8126	650	127.0
	Random	64	1627	403	25.4
	GHZ	64	63	66	1.0
	Graphstate	64	64	25	1.0
	QPEexact	64	2139	626	33.4
	QAOA	64	3920	493	61.2
	Multiplier	64	13040	22469	203.8

H-grid: six 4×4 cores (96 physical qubits, 7 EPR links). See Figure 2 for the full qubit-level topology. The 25q suite on the B-grid and the 64q suite on the H-grid follow the benchmark setup of TelesABRE [?] (we add a 36q B-grid suite to broaden circuit diversity).

Baselines. We compare against two state-of-the-art distributed routers: (i) TelesABRE [?], a SABRE-style router that jointly optimises intra-core SWAPs, teledata (qubit moves), and telegate (remote CX execution) operations; and (ii) pytket-dqc [?], a Quantinuum library that distributes circuits via *gate teleportation* with multi-gate amortisation using KaHyPar hypergraph partitioning. TelesABRE is run using its compiled C binary with the `optimize_initial` flag enabled (built-in layout heuristic + 3-success sampling) on the same device; pytket-dqc uses its HypergraphPartitioning allocator (best of 5 random seeds; its hypergraph encoding requires a minor weight-0 patch to run on the KaHyPar $\geq 1.3.5$ release we use). Internally, we also compare against dSABRE-Topo, an ablation that replaces the BFS-layer extended set with a topological-order one (Section ??).

What one EPR pair buys. All routers report in the same unit (one EPR pair = one e-bit; “EPR” and “e-bit” are used interchangeably), but what each unit *buys* differs, and no cross-compiler EPR comparison is meaningful without stating the difference. Table ?? summarises the four cost models in play. dSABRE uses *teledata*: one EPR per inter-core hop relocates a logical qubit, after which subsequent intra-core gates on it are free. pytket-dqc uses *gate teleportation*: one EPR distributed along a Steiner tree of cores executes a whole hyperedge of CNOTs sharing one control, paying one EPR per Steiner edge rather than per CNOT. TelesABRE uses both primitives and we count each operation once. Circuits with isolated inter-core gates favour teledata; circuits with long shared-

TABLE IV
WHAT ONE EPR PAIR BUYS UNDER EACH COMPILER’S COST MODEL.
COUNTS ARE UNIT-CONSISTENT ACROSS ROWS BUT PURCHASE
DIFFERENT AMOUNTS OF WORK, AND ONLY THE FIRST TWO ROWS
CHARGE INTRA-CORE SWAPS.

Compiler	Primitive	One EPR pair buys	SWAPs
dSABRE	teledata	one qubit relocation, one hop	counted
TelesABRE	teledata + telegate	a relocation, or one remote CX	counted
pytket-dqc	gate teleport	one Steiner edge of a hyperedge	not modelled
DMapS	remote SWAP	half a two-qubit exchange (2 EPR)	counted

TABLE V
HARDWARE COST AND HEURISTIC PARAMETERS

Parameter	Symbol	Value
SWAP cost	c_{swap}	3
Teleport (EPR) cost	c_{tele}	10
Capacity threshold	τ	3
Capacity penalty	c_{pen}	15
Inter-core link weight	w_{link}	10
Extended-set weight	w_e	0.25
Extended-set capacity	L	20
Lookahead decay	γ	0.9
Deadlock limit	L_{deadlock}	50
Max rollbacks	$N_{\text{backup}}^{\text{max}}$	50

control CNOT runs (QFT, QPE) favour gate teleportation.

Three consequences shape how the tables below should be read. (i) *Work per unit differs*, so the Δ_{tket} columns in Tables ??–?? are unit-consistent but not work-consistent. (ii) *Scope differs*: pytket-dqc emits an e-bit count only and does not model intra-core SWAP routing at all, so its column omits a cost dSABRE pays and reports. (iii) The asymmetry is also *temporal*—a gate-teleportation e-bit must stay entangled for the full span of the hyperedge it serves, whereas a teledata e-bit is consumed instantaneously by its teleport—so pytket-dqc’s counts presuppose an entanglement lifetime that hardware may not provide. We address (i) and (ii) with a layout-controlled comparison in which both compilers start from the same qubit-to-core partition, and (iii) with a lifetime sweep; both are reported in Section ?. Against TelesABRE, which shares dSABRE’s cost model and reports intra-core SWAPs, no such correction is needed and the comparison is direct.

Parameters. Table ?? lists all hyperparameters. dSABRE is deterministic given a fixed layout; we run it with three SabreLayout seeds and report the best EPR count (lower = better). TelesABRE is stochastic and is run with three random seeds; again the best result is reported, following established practice [?]. Layout passes: 2 (forward–backward–forward).

On best-of- N reporting. Each router uses its authors’ convention: best-of-3 seeds for dSABRE and TelesABRE [?], best-of-5 for pytket-dqc [?]. This reflects the operational cost rather than per-attempt variance. The advantage over TelesABRE is robust to this choice: replacing best with median on the original five-circuit TelesABRE-matched 64q set shifts the gmean gap from -52.6% to -50.2% vs. TelesABRE, still a large reduction. The comparison against pytket-dqc is method-dependent (Tables ??–??) and is discussed per suite above.

TABLE VI

25-QUBIT SUITE (B-GRID $2 \times 2 \times 4 \times 4$, 64 PHYSICAL QUBITS). EPR = TELEDATA + TELEGATE (1 EPR PAIR EACH) FOR TELESABRE, TELEDATA FOR DSABRE, E-BITS FOR PYTKET-DQC; SWAP COUNTS INTRA-CORE SWAPS ONLY (PYTKET-DQC DOES NOT MODEL INTRA-CORE ROUTING). Δ_{TS} AND Δ_{tket} ARE DSABRE'S EPR CHANGE RELATIVE TO TELESABRE AND PYTKET-DQC RESPECTIVELY; NEGATIVE VALUES INDICATE A REDUCTION IN EPR COST BY DSABRE (LOWER IS BETTER). TELESABRE NUMBERS ARE BEST-OF-3-SEED; PYTKET-DQC USES ITS STRONGEST DISTRIBUTOR, COVEREMBEDDINGSTEINERDETACHED (STEINER-TREE GATE EMBEDDING WITH DETACHED GATES), BEST-OF-5-SEED; DSABRE RUNS WITH THE DEFAULT CONFIGURATION DESCRIBED ABOVE. COST ANALYSIS IS DEFERRED TO SECTION ??.

Circuit	CX	TelesABRE		dsABRE		tket	Δ EPR (%)	
		EPR	SWAP	EPR	SWAP	e-bits	TS	tket
ae	558	23	297	23	284	48	+0.0	-52.1
ghz	24	2	29	1	6	3	-50.0	-66.7
graphstate	25	11	51	2	15	4	-81.8	-50.0
qft	580	39	355	33	334	61	-15.4	-45.9
qnn	1223	51	587	48	663	115	-5.9	-58.3
random	1124	292	1273	178	1241	479	-39.0	-62.8
gmean		25.8	221	15.3	138	35.3	-40.6	-56.6

B. Main Results

TelesABRE [?] supports three operation types: intra-core SWAPs, teledata (qubit movement, 1 EPR), and telegate (remote CX execution via the cat-entangler protocol, 1 EPR). We report TelesABRE's EPR count as the sum of teledata and telegate operations, so each EPR pair is counted exactly once. Unless stated otherwise, dsABRE is run with its default initial-mapping configuration: Qiskit SabreLayout on the corners-removed architecture graph, best of three seeds, followed by SABRE-style forward→backward→forward routing passes (Section ??). Subsequent tables and ablations inherit this default. We report two metrics in the main tables: EPR pairs and intra-core SWAPs. Cost analysis is deferred to Section ??.

25q. dsABRE reduces geometric-mean EPRs over TelesABRE by **40.6%** and over pytket-dqc by **56.6%** (Table ??). Per-circuit reductions versus TelesABRE range from parity on AE to -81.8% on Graphstate, with no regressions on this suite. Even against pytket-dqc's strongest Steiner-embedding distributor, dsABRE wins on all six circuits (-45.9 to -66.7%): state teleportation amortises a single EPR pair over many subsequent local gates, and at this scale that outweighs the gate-embedding savings pytket-dqc extracts by reusing shared control qubits.

36q. dsABRE reduces geometric-mean EPRs by **46.3%** vs. TelesABRE and **12.3%** vs. pytket-dqc (Table ??). Per-circuit reductions versus TelesABRE are large (-35% to -80%, DJ at parity). The simple structured circuits BV, VQE-SU2, and W-state are essentially solved (1 EPR on BV; 6-10 EPRs on the others). Against pytket-dqc's Steiner-embedding distributor the picture is mixed: dsABRE matches or beats it on four circuits (BV -66.7%, QAOA -7.4%; DJ and W-state at parity), while the embedding is more e-bit-efficient on QPEexact (+32.7%) and VQE-SU2 (+11.1%), whose interaction graphs KaHyPar partitions almost perfectly; the geometric mean still favours dsABRE.

64q. The H-grid suite is where the gap from TelesABRE

TABLE VII

36-QUBIT SUITE (SAME B-GRID AS 25-QUBIT SUITE). METHODOLOGY AND NOTATION AS TABLE ??.

Circuit	TelesABRE			dsABRE		tket	Δ EPR (%)	
	CX	EPR	SWAP	EPR	SWAP	e-bits	TS	tket
bv	17	5	34	1	12	3	-80.0	-66.7
dj	35	3	53	3	23	3	+0.0	+0.0
qaoa	1200	232	1027	137	1015	148	-40.9	-7.4
qpeexact	1019	100	771	65	586	49	-35.0	+32.7
vqe_su2	105	16	80	10	56	9	-37.5	+11.1
wstate	70	12	88	6	25	6	-50.0	+0.0
gmean		20.1	147	10.8	78	12.3	-46.3	-12.3

is most consequential (Table ??). The suite comprises the six TelesABRE-benchmark circuits plus the three class-completion additions (QPEexact, QAOA, Multiplier). dsABRE reduces gmean EPRs by **45.5%** over the seven circuits on which TelesABRE converges, with per-circuit reductions ranging from -12.6% (QPEexact) to -78.9% (Graphstate), QFT/QNN at -50.2%/-62.8%, and the 13k-CX Multiplier at -31.2%. The per-core adaptive corner reservation (Section ??) supplies every core with escape slots, which resolves the earlier star-circuit weakness on GHZ: it now improves (9 vs. 11) rather than regressing, as no core is born fully packed and the local Δ -form score is no longer forced to chase the next CX out of a saturated core. Against pytket-dqc's strongest distributor the comparison *reverses*: the 9-circuit geometric mean is +20.0%, i.e. dsABRE's teleport-EPRs exceed pytket-dqc's e-bits. COVEREMBEDDINGSTEINERDETACHED is markedly more e-bit-efficient on the structured circuits (AE +71.6%, GHZ +80.0%, Graphstate +77.8%, QFT +12.7%, and above all QPEexact +182.3%; QAOA is near parity at +12.1%), where Steiner-tree gate embedding reuses a few EPR pairs across many controlled operations; dsABRE leads on the three densest circuits (QNN -30.9%, Random -25.5%, Multiplier -48.8%), on which the embedding method does not scale or embed and a plain partitioner is used. Two caveats temper this: pytket-dqc's e-bit count omits the intra-core SWAP cost that dsABRE pays and reports, and its strongest method is computationally far heavier (minutes to intractable per large circuit versus dsABRE's seconds). On Random and QAOA, TelesABRE fails to converge on any of three seeds; dsABRE completes both (721 and 529 EPR) under default scoring and proactive relief alone, without triggering checkpoint-rollback.

a) *Shared-mapping comparison.*: To isolate routing from allocation quality, we re-run dsABRE from each baseline's own initial layout. The TelesABRE gap persists (-25% EPR on 25q, -43% on 64q from its optimize_initial layout); against pytket-dqc's KaHyPar partition dsABRE wins on all three suites (-56.8% on 25q, -8.0% on 36q, -40.1% on 64q); the 36q margin is the narrowest because the sparse interaction graphs in that suite are already near-optimally partitioned by KaHyPar.

b) *Entanglement-lifetime sensitivity.*: Every pytket-dqc e-bit count above assumes an *unbounded* entanglement lifetime: the link qubit created by one starting

TABLE VIII

64-QUBIT SUITE (H-GRID $2 \times 3 \times 4$, 96 PHYSICAL QUBITS). METHODOLOGY AND NOTATION AS TABLE ??; *TELESABRE FAILS TO CONVERGE ON RANDOM AND QAOA. $\dagger$ PYTKET-DQC'S COVEREMBEDDINGSTEINERDETACHED DOES NOT SCALE TO (QNN, MULTIPLIER) OR EMBED (RANDOM) THESE DENSE CIRCUITS WITHIN A 20-MINUTE BUDGET; THE NON-EMBEDDING PARTITIONINGHETEROGENEOUS DISTRIBUTOR IS USED FOR THOSE CELLS.

Circuit	CX	TeleSABRE		dsABRE		tket	Δ EPR (%)	
		EPR	SWAP	EPR	SWAP	e-bits	TS	tket
ae	1962	323	1508	242	1493	141	-25.1	+71.6
ghz	63	11	111	9	57	5	-18.2	+80.0
graphstate	64	76	379	16	89	9	-78.9	+77.8
qft	1966	410	1872	204	1563	181	-50.2	+12.7
qnn	8126	1365	6292	508	5229	735 $\dagger$	-62.8	-30.9
random*	1627	—	—	721	3608	968 $\dagger$	—	-25.5
qpeexact	2139	310	1624	271	1792	96	-12.6	+182.3
qaoa*	3920	—	—	529	3688	472	—	+12.1
multiplier	13040	2202	11232	1514	10109	2957 $\dagger$	-31.2	-48.8
gmean (7)		269.5	1452	147.0	1016	113.3	-45.5	+29.7
gmean (9)		—	—	202.2	1350	168.5	—	+20.0

process stays entangled until the last gate of its hyperedge has executed. Bandini *et al.* [?] argue that this is unrealistic—EPR-pair quality decays quickly, so the number of gates sharing one pair must be bounded—and cap gates-per-EPR at $D_{\max}$, proposing $D_{\max}=3$ as a realistic setting. We therefore re-score pytket-dqc's distributions under this bound, using a counter that generalises its own cost model: an established link serves at most $D_{\max}$ gates before it dies and must be re-established from a still-live neighbour on the hyperedge's Steiner tree. The counter is conservative in pytket-dqc's favour (intermediate links age only when they themselves serve gates; embedded and home-server gates age nothing; wall-clock time is ignored) and reproduces pytket-dqc's own cost exactly at $D_{\max}=\infty$ on every distribution evaluated. dsABRE needs no adjustment at any $D_{\max}$: a teledata EPR is consumed instantaneously by its teleport, so teledata counts are lifetime-invariant by construction.

Table ?? shows the sweep on the 64q suite. The +20.0% reversal of Table ?? exists only above a lifetime threshold: the geometric-mean gap crosses parity between $D_{\max}=10$ and $D_{\max}=5$ and becomes a dsABRE win of **37.9%** at the realistic $D_{\max}=3$ (68.5% at $D_{\max}=1$). The circuits pytket-dqc wins at $D_{\max}=\infty$ are exactly those whose distributions hold links across many gates, and they inflate $2.7\text{--}4.8\times$ at $D_{\max}=3$; the sharpest case is QPEexact, its largest unbounded win (+182.3%), which inflates $79 \rightarrow 378$ e-bits and flips to a dsABRE advantage of 28.3%. Its winning QNN distribution requires individual link qubits to stay entangled across more than half the circuit's execution. Circuits whose hyperedges hold roughly one gate each (GHZ, Graphstate) are lifetime-insensitive, so those residual wins reflect partition quality rather than held entanglement—consistent with the shared-mapping result above—and the dense circuits, which dsABRE already led, move least. The smaller suites behave the same way, only more so: QPEexact-36, the one structured 36q circuit pytket-dqc won, becomes a 66% dsABRE advan-

TABLE IX

ENTANGLEMENT-LIFETIME SENSITIVITY ON THE 64Q SUITE: PYTKET-DQC E-BITS WHEN ONE EPR PAIR MAY SERVE AT MOST $D_{\max}$ GATES (BEST COMPLETED DISTRIBUTOR PER CELL, COUNTER AS DESCRIBED IN THE TEXT). THE $D_{\max}=\infty$ COLUMN REPRODUCES THE SETTING OF TABLE ?? UP TO SEED AND PIPELINE VARIATION AND IS IN AGGREGATE *more* FAVOURABLE TO PYTKET-DQC (GMEAN +33.4% VS +20.0%). dsABRE EPR (TABLE ??) IS $D_{\max}$ -INVARIANT. Δ_3 COMPARES dsABRE AGAINST THE $D_{\max}=3$ COLUMN; NEGATIVE FAVOURS dsABRE.

Circuit	dsABRE EPR	pytket-dqc e-bits at $D_{\max}$					Δ_3 (%)
		∞	10	5	3	1	
ae	242	137	191	292	423	1118	-42.8
ghz	9	5	5	5	5	5	+80.0
graphstate	16	6	6	6	6	6	+166.7
qft	204	190	265	400	572	1416	-64.3
qnn	508	540	1009	1592	2435	6866	-79.1
random	721	908	908	922	967	1453	-25.4
qpeexact	271	79	136	245	378	1064	-28.3
qaoa	529	472	548	878	1282	3402	-58.7
multiplier	1514	2950	3352	4008	5004	10930	-69.7
gmean Δ (%)		+33.4	+5.6	-20.5	-37.9	-68.5	

TABLE X

ABLATION OF THE INTER-CORE EXTENDED-SET CONSTRUCTION. BOTH VARIANTS ARE THE SAME ROUTER DIFFERING ONLY IN HOW THE INTER-CORE E IS BUILT: TOPOLOGICAL ORDER VS. BFS-LAYER EXPANSION (DEFAULT). GEOMETRIC MEAN OVER EACH SUITE.

Suite	dsABRE-topo		dsABRE		Δ (%)	
	EPR	SWAP	EPR	SWAP	EPR	SWAP
25q	16.4	141	15.3	138	-6.8	-2.1
36q	12.1	94	10.8	78	-10.9	-16.5
64q	127.4	832	117.3	779	-7.9	-6.3

tage at $D_{\max}=3$. A lifetime-aware re-partition could soften this inflation, but pytket-dqc offers no such option. We conclude that gate teleportation's e-bit advantage is conditional on hardware sustaining entanglement across $\gtrsim 5\text{--}10$ gates per pair, whereas teledata routing delivers its counts under any lifetime. Per-circuit figures are in the online appendices.

C. Ablation Study

We conduct a systematic ablation of dsABRE's design choices across three axes: (i) the extended-set construction, (ii) discrete mechanism on/off, and (iii) continuous parameter sensitivity. All ablations inherit dsABRE's default configuration (Section ??) and report the *geometric-mean EPR count* (gmEPR) over the respective suite, so the dsABRE / **Full** baselines in Tables ?? and ?? match the headline gmEPR in Tables ????

a) *Inter-core extended-set construction: BFS vs. topological.*: To isolate the contribution of the BFS-layer inter-core extended set (Section ??) we substitute the standard topological-order construction in the teleport scorer (the intra-core extended set is unchanged) and rerun all three suites. Table ?? reports the result.

BFS reduces gmean EPRs by 6.8%, 10.9%, and 7.9% on the three suites respectively. The gain is largest on the 36q suite and reflects the richer dependency structure in larger circuits,

TABLE XI

MECHANISM ABLATION ON THE 25Q AND 64Q SUITES. ONE COMPONENT IS DISABLED PER ROW; ALL OTHERS REMAIN AT THEIR DEFAULTS. SAME PROTOCOL AS THE MAIN RESULTS, SO THE **FULL** BASELINE MATCHES THE HEADLINE GMEPR IN TABLES ?? AND ??.

Configuration	25q		64q	
	gmEPR	Δ (%)	gmEPR	Δ (%)
Full (baseline)	15.3	—	117.3	—
No extended-set lookahead ($w_e = 0$)	17.1	+11.3	148.3	+26.5
No capacity penalty ($c_{\text{pen}} = 0$)	24.7	+60.8	180.0	+53.5
No hop-gain reward ($w_h = 0$)	15.3	0.0	119.5	+1.8
No congestion relief	15.3	0.0	107.1	-8.7

where topological order drifts further from the active front layer. SWAP counts fall on all three suites (-2.1% , -16.5% , -6.3%); the BFS scorer resolves the front through better-placed teleportations and leaves less residual intra-core work.

b) Mechanism ablation.: Table ?? ablates each scoring component in isolation on the 25q and 64q suites by setting the corresponding parameter to zero or disabling the mechanism entirely. The two suites stress different regimes: the 25q B-grid is sparsely loaded (qubits per core ≤ 7 , ample headroom on every core), so congestion-driven mechanisms barely activate, while the 64q H-grid runs at $\sim 67\%$ qubit utilisation and saturates inter-core ports under dense workloads.

The capacity penalty is the single most critical component on both suites: without it the router greedily teleports into full cores, triggering cascading evictions that inflate gmean EPR by $\sim 61\%$ on 25q and $\sim 54\%$ on 64q. The extended-set lookahead contributes $+11\%$ on 25q and $+27\%$ on 64q; removing it collapses the scorer to a near-myopic policy, and its value grows with circuit size as dependency chains lengthen. The hop-gain reward — a directional bonus for moves that reduce inter-core hop count — is inert on the 25q B-grid (where the front-layer weighted distance already captures direction) but contributes a small improvement on the larger 64q H-grid ($+1.8\%$ gmean): longer inter-core paths mean the specific landing port can be far from the optimal staging position for the next hop, so Δ_F understates the true directional benefit (it is less negative than the move deserves); g_{hop} compensates with a fixed position-independent signal.

Congestion relief is near-neutral on both suites once the per-core adaptive layout is in place: it is inactive on 25q (0%) and mildly counterproductive on 64q (-8.7% , i.e. disabling it lowers gmean EPR). Because every core now begins with reserved escape capacity, the standing congestion the relief mechanism was designed to pre-empt is largely absorbed upstream by the layout; its residual value is confined to individual hard instances (e.g. 64q Random, where it enables completion) rather than the suite geometric mean.

c) Parameter sensitivity.: We sweep each continuous hyperparameter one-at-a-time on 25q and 64q. All defaults— $w_e=0.25$, $c_{\text{pen}}=15$, $|E|=20$, $\gamma=0.9$, $w_{\text{link}}=10$, and the relief bonus—sit on a flat region of their respective curves (within $\sim 2\%$ of the local minimum), so performance is robust to small perturbations. The only sharp regressions occur when a control is pushed far from its default: e.g. $w_e \geq 1.0$ inflates

25q gmEPR by $3\text{--}5\times$ as the lookahead dominates immediate preparation costs, $c_{\text{pen}}=0$ doubles gmEPR by inviting deadlocks in saturated cores, and $\gamma=1.0$ (flat weighting) costs 7% on 64q. These observations confirm that the defaults are well-calibrated and that no single hyperparameter is a hidden lever; tuning beyond the defaults yields diminishing returns.

D. SabreLayout-Based Initial Layout

dSABRE bootstraps from Qiskit SabreLayout [?] run on the full architecture graph G_A (intra-core edges plus inter-core links), with two adaptations for the distributed setting.

Per-core corner reservation. SabreLayout packs each core to its full physical capacity. Two problems follow. First, an occupied communication port cannot accept an incoming teleport without first evicting its current occupant, so saturated cores either stall the inter-core scorer or force chains of relocations. Second, full cores trip the capacity penalty (Section III-D) on every candidate teleport into them, biasing the router away from the actually-preferred landing core. We therefore exclude, from the coupling map passed to SabreLayout, the k most-remote corner physical qubits of every core—the minimum-degree vertices farthest (by total shortest-path distance) from all other physical qubits, which in both B-grid and H-grid architectures sit furthest from any inter-core link—so that each core retains k free slots for eviction scratch space and stays below the capacity threshold. The count k is chosen adaptively as the largest value in $\{0, \dots, 4\}$ that keeps the usable-slot fill $n_q/(N_{\text{cores}}(m^2 - k))$ at or below 0.80; this yields $k=4$ on the 25/36/100/200-qubit suites and $k=2$ on the denser 64-qubit suite. A whole-chip variant that removes only the globally lowest-degree corners reserves slots in a single core under core-major qubit numbering, leaving the others fully packed; per-core reservation is what supplies every core with escape capacity.

Forward-backward-forward schedule. We run three routing passes (forward $\rightarrow$ reversed-DAG backward $\rightarrow$ forward) and keep the better of the two forward passes by EPR. The backward pass starts from the first forward pass’s output layout ℓ_f and routes the reversed DAG, producing ℓ_b that pre-positions qubits near their early interaction partners; the third pass then warm-starts from ℓ_b . This is the same mechanism as SABRE’s backward refinement [?] and outperforms pure-forward warm-restart. Best of three SabreLayout seeds is reported.

E. Compile Time

Table ?? reports wall-clock compilation time on the 64-qubit suite.

Times scale with CX count: TeleSABRE spans 0.02 s (GHZ) to 86 s (Multiplier, 13 040 CX); dSABRE spans 0.05 s to 462 s. dSABRE is slower on most circuits where both converge. The gap is an implementation constant, not an asymptotic one, and three pieces of evidence separate the two. First, the measured growth matches the complexity of Eq. ?? rather than exceeding it: dSABRE’s time grows with CX count at the same rate as TeleSABRE’s, and the ratio between

TABLE XII

COMPILATION TIME IN SECONDS (64-QUBIT SUITE, H-GRID $2 \times 3 \times 4$). TELESABRE: BEST-SEED WALL-CLOCK TIME. DSABRE: TOTAL TIME FOR THREE ROUTING PASSES (FWD→BWD→FWD) FROM THE BEST SABRELAYOUT SEED. TELESABRE IS A COMPILED C++ BINARY; DSABRE IS PURE-PYTHON WITH NETWORKX. *TELESABRE FAILS TO CONVERGE ON RANDOM AND QAOA.

Circuit	CX	TelesABRE (s)	dsABRE (s)
AE	1962	1.05	12.60
GHZ	63	0.02	0.09
Graphstate	64	3.92	0.05
QFT	1966	1.81	10.34
QNN	8126	25.21	123.60
Random*	1627	—	16.69
QPEexact	2139	4.87	8.58
QAOA*	3920	—	26.63
Multiplier	13040	85.69	462.44

them stays bounded—within about one order of magnitude—across a $200\times$ range of circuit size, which is the signature of a per-operation overhead rather than a worse algorithm. Second, the two implementations differ in exactly the way that constant predicts: TelesABRE is a compiled C++ binary with cached distance lookups, while dsABRE is pure Python calling NetworkX for shortest paths inside the routing loop, and interpreted-versus-compiled overheads of one order of magnitude are the norm for this kind of inner loop. Third, the ordering inverts whenever the constant stops dominating: on Graphstate dsABRE is $78\times$ faster (0.05 s vs. 3.92 s), because TelesABRE’s three-success sampling spends many iterations before three runs converge there. Nothing in Eq. ?? is specific to Python, so a compiled implementation would be expected to close most of the gap; we regard that as engineering rather than algorithm design and have not pursued it. In absolute terms the cost is modest against what it buys: dsABRE routes the largest circuit in the suite in under eight minutes and roughly halves its EPR count, and EPR pairs—not compile seconds—are the scarce resource on the hardware this targets.

F. Large-Circuit Scalability

To stress-test dsABRE substantially beyond the main evaluation range we run dsABRE, TelesABRE, and pytket-dqc on three additional suites using the MQT-Bench QFT circuit (IBM native-gate, Qiskit opt3). The architectures scale the H-grid of the 64q suite:

- **100q:** H-grid $2 \times 3 \times 5 \times 5$ (150 physical).
- **200q:** H-grid $4 \times 3 \times 5 \times 5$ (300 physical).
- **360q:** H-grid $2 \times 3 \times 9 \times 9$ (486 physical).

TelesABRE is best-of-3-seed with a 600 s per-seed timeout for 200q+; dsABRE uses its default configuration; pytket-dqc uses HypergraphPartitioning (best of 5 seeds). Each router runs from its own initial layout, so the Δ_{tket} column should be read with this layout asymmetry in mind in addition to the e-bit counting caveat (Section ??).¹

¹For a layout-controlled point of comparison, re-running dsABRE from pytket-dqc’s HypergraphPartitioning layout gives EPR counts of 280 (100q), 1014 (200q), and 563 (360q), versus pytket-dqc’s 815, 1124, and 669 — corresponding to Δ_{tket} of -65.6% , -9.8% , and -15.8% . In this matched-layout setting dsABRE beats pytket-dqc at all three sizes, most narrowly at 200q where its SabreLayout configuration is marginally worse.

TABLE XIII

LARGE-CIRCUIT SCALABILITY (QFT). EACH ROUTER RUNS FROM ITS OWN INITIAL LAYOUT. NEGATIVE ΔEPR IS BETTER. TELESABRE FAILS TO CONVERGE ON THE 200Q CIRCUIT WITHIN THE 600 s TIMEOUT. [†]SEE CAVEAT ON E-BIT COUNTING. [‡]PYTKET-DQC’S COVEREMBEDDINGSTEINERDETACHED EXCEEDS A 25-MINUTE BUDGET AT 200Q AND 360Q; THE PARTITIONINGHETEROGENEOUS DISTRIBUTOR IS USED FOR THOSE E-BIT COUNTS.

Suite	CX	TelesABRE		dsABRE		tket [†] e-bits	ΔEPR (%)	
		EPR	SWAP	EPR	SWAP		TS	tket
100q	3420	248	2414	223	2723	114	-10.1	+95.6
200q	7220	—	—	834	7043	447 [‡]	—	+86.6
360q	13300	1109	15098	626	13433	195 [‡]	-43.6	+221.0

Table ?? shows the QFT scalability trend. At 100q, dsABRE now edges TelesABRE (-10.1% EPR under the per-core adaptive layout), but pytket-dqc’s Steiner-tree embedding is far more e-bit-efficient on this deep controlled-phase chain ($+95.6\%$: dsABRE uses 223 teleport-EPRs versus 114 e-bits). At 200q, TelesABRE fails to converge within the 600 s timeout, whereas dsABRE completes (834 EPR) under its default scoring and proactive relief; the checkpoint-rollback escape (Section ??) does not fire on the best-EPR run, but is what salvages one of the three SabreLayout seeds that would otherwise abort, leaving the best-of-three selection well-fed. The embedding advantage persists here ($+86.6\%$ vs. pytket-dqc), though COVEREMBEDDINGSTEINERDETACHED exceeds a 25-minute budget and the cheaper PARTITIONINGHETEROGENEOUS distributor is reported instead. At 360q, dsABRE again beats TelesABRE (626 vs. 1109 EPR, -43.6%) while pytket-dqc’s embedding is more e-bit-efficient (195 e-bits, $+221\%$); the per-core adaptive layout together with the removal of the node-decay term cuts dsABRE’s SWAP count from 27,489 to 13,433 (-51%) relative to the previous configuration. As at the smaller sizes, the QFT comparison against pytket-dqc is dominated by its gate-teleportation embedding on this circuit family, which dsABRE’s state-teleportation model does not target; the EPR-versus-e-bit caveat (no intra-core SWAP cost for pytket-dqc) applies throughout. Compile times for 360q QFT are 281 s (TelesABRE, C++, best of 3 seeds) and, for dsABRE (Python, SabreLayout + fwd→bwd→fwd), of order 700 s per SabreLayout seed; the Python overhead is the primary driver of dsABRE’s longer wall time relative to the compiled C++ TelesABRE. pytket-dqc’s strongest distributor is the most expensive of the three—COVEREMBEDDINGSTEINERDETACHED exceeds a 25-minute budget at this size (its cheaper HYPERGRAPH-PARTITIONING partitioner runs in ≈ 550 s).

G. Cost Analysis

The combined-cost ratio $c_{\text{tele}}:c_{\text{swap}}$ is hardware-dependent: near-term microwave-coupled modular superconducting qubits keep teleportation within a small constant factor of local SWAPs [?], while photonic-link architectures have measured EPR-generation times three to four orders of magnitude slower than local gates [?], [?], pushing the ratio towards 100:1 or beyond. Table ?? sweeps c_{tele} from 10 to 100 with

TABLE XIV
COST-MODEL SENSITIVITY: GEOMETRIC-MEAN COST REDUCTION OF
DSABRE OVER TELESABRE AS c_{tele} VARIES ($c_{\text{swap}} = 3$ FIXED).
NEGATIVE IS BETTER. 64Q EXCLUDES THE RANDOM CIRCUIT ON WHICH
TELESABRE ABORTS.

c_{tele}	25q	36q	64q
10	−39.2	−47.0	−44.6
20	−39.8	−47.0	−46.9
50	−40.3	−47.0	−49.5
100	−40.5	−46.8	−50.9

$c_{\text{swap}} = 3$ held fixed, reporting geometric-mean cost reduction of dSABRE vs. TeLESABRE.

The pattern is consistent across all three suites: the gap to TeLESABRE widens as c_{tele} grows. At the teleport-friendly end ($c_{\text{tele}}=10$, ratio $c_{\text{tele}}/c_{\text{swap}} \approx 3.3$) the cost model penalises teleports least, so TeLESABRE’s higher EPR count is discounted; dSABRE still saves 20.6% on 25q, 39.0% on 36q, and 23.8% on 64q. At $c_{\text{tele}}=100$ (ratio ≈ 33) — the SWAP-favourable regime of near-term photonic-link hardware [?], [?] — dSABRE’s smaller EPR count dominates and the savings reach 35.2%, 43.2%, and 38.9% respectively. Across the entire practically relevant range dSABRE dominates TeLESABRE, and the advantage grows with EPR cost — exactly the regime where the cost model becomes stricter.

V. RELATED WORK

The Introduction already places dSABRE within the partition–communicate–schedule decomposition of the DQC compilation stack and lists the major lines of work in each stage. Here we focus on what specifically distinguishes dSABRE from the closest prior art and on the single-QPU SABRE family that we build on but do not subsume.

SABRE-style distributed routers. TeLESABRE [?] is our closest prior work and direct experimental baseline. Both routers follow the SABRE loop, but differ in how they score inter-core moves: TeLESABRE evaluates candidates via a *contracted communication-graph energy* that averages per-gate Dijkstra distances over the whole front layer (extended in topological order), whereas dSABRE uses a local, gate-centric five-term score (Eq. ??) that explicitly accounts for intra-core staging cost, core capacity, and directional progress, paired with a BFS-layer extended set (Section ??). This makes dSABRE’s inter-core scoring directly comparable in form to its intra-core SWAP scoring (Eq. 5). dSABRE additionally provides proactive congestion-relief candidates (Section ??) and a checkpoint–rollback deadlock escape (Section ??), whereas TeLESABRE relies on a safety-valve iteration cap; TeLESABRE supports telegate candidates while dSABRE currently uses teledata only.

DMapS [?] is a concurrent end-to-end DQC compiler that pairs a KaHyPar partitioner with a SABRE-derived router and shares our joint EPR-plus-local-SWAP objective. Two design choices set it apart from dSABRE: its only cross-core primitive is a two-EPR *remote SWAP* (whereas dSABRE uses one-EPR one-way teledata into a layout-reserved free slot), and its initial allocation comes from a KaHyPar partition of the interaction

hypergraph (whereas dSABRE uses SabreLayout on $G_{\mathcal{A}}$). On the dense circuits in our suites dSABRE consumes substantially fewer EPRs — e.g. $2\text{--}4\times$ on 25q QFT/QNN/Random and $1.6\text{--}1.8\times$ on 64q QFT/QNN/Random — while DMapS wins on maximally sparse circuits (GHZ, Graphstate, W-state) where KaHyPar partitions interaction graphs almost perfectly. A code-level audit, the full per-circuit head-to-head, and a controlled fill-ratio sweep showing parity once the device fill ratio approaches 1 are included in the online appendices accompanying the paper (see “Artefact availability” below).

Allocation methods built on hypergraph partitioning [?], [?] are complementary to dSABRE: any can supply the initial layout its routing loop consumes. Similarly, time-aware schedulers [?], [?] sit upstream or downstream of the routing stage dSABRE addresses.

Single-QPU qubit mapping. On a single-core processor the routing problem reduces to inserting intra-chip SWAPs. The closest single-QPU antecedent to our work is the front+extended-set scorer SABRE [?] that we and TeLESABRE both build on. Babu et al. [?] use auxiliary-qubit gate teleportation on a heavy-hex IBM Eagle processor, achieving 10–25% depth reduction versus standard SABRE; the setting is single-QPU and not directly comparable to multi-core routing.

VI. FUTURE WORK

Composing dSABRE with burst execution. dSABRE optimises EPR count on a fixed circuit DAG. Burst-execution passes such as AutoComm [?] and its collective extension CollComm [?] operate earlier in the toolchain: they use gate commutativity to reorder consecutive non-local gates on the same wire (or across multiple wires) and amortise the whole block onto a single shared entanglement state, reporting EPR savings of 50–75% on amenable circuits; [the lifetime-bounded gate grouping of Bandini et al. \[?\] tempers such savings with the \$D_{\text{max}}\$ constraint we adopt in Table ??](#). The two approaches operate at different layers and should compose well: a burst-extraction pre-pass would expose longer wire-aligned gate sequences, and dSABRE’s BFS extended set would let the teleport score capture their full benefit. Quantifying the joint improvement—especially on circuits where a burst’s beneficiary core is itself a function of the routing decisions dSABRE makes—is an open question, and recent integrations of gate-teleportation amortisation with SABRE-style mappers [?] suggest the interaction is non-trivial.

Parallel teleportation for circuit-runtime minimisation. dSABRE optimises EPR count and emits one teleport per iteration; wall-clock runtime depends additionally on whether teleports with disjoint links and staging sets can fire concurrently. A natural follow-on is a scheduling pass over dSABRE’s teleport trace that batches non-conflicting moves, co-designed with denser inter-core links [?] and entanglement buffer qubits [?], [?]. Time-sliced compilers [?], [?] provide a starting point but are not paired with a SABRE-style scoring loop; naively adding links without retuning the routing energy can *increase* EPR consumption.

VII. CONCLUSION

We presented dSABRE, a SABRE-style router for multi-core distributed quantum computers that, on each iteration of a lookahead-driven loop, resolves intra-core front-layer gates by SWAP scoring and only scores inter-core teleport candidates when the intra-core front is empty, with both phases sharing a common decay-weighted lookahead form. The empirical takeaway is that three design choices — an explicit capacity penalty that keeps the scorer from teleporting into saturated cores, proactive congestion relief triggered by an explicit demand vector, and a BFS-layer inter-core extended set that respects DAG dependencies — account for most of the gap to the state of the art, both individually and together, and that this gap widens as circuits and architectures scale. More broadly, the results indicate that distributed routing benefits less from a richer per-candidate scoring formula than from ensuring that the lookahead signal and the architectural capacity constraints enter the same scoring step.

Artefact availability.: The dSABRE implementation, the routed-circuit benchmark suites, per-circuit JSON results, and the online appendices referenced throughout this paper are available at <https://github.com/ebony72/dsabre>.

ACKNOWLEDGMENTS

This work was partially supported by the Australian Research Council (Grant No. DP250102952 and DP220102059). The author used Anthropic’s Claude (via Claude Code) to assist with: (i) refactoring portions of the dSABRE implementation, (ii) authoring benchmark scripts and aggregating per-circuit JSON results into the tables of Section ??, (iii) drafting the LaTeX sources for the architecture and teleport-candidate figures (Figures 2 and ??), and (iv) copy-editing prose throughout the manuscript. All AI-generated content was reviewed, verified, and corrected by the author, who takes full responsibility for the correctness and originality of the work.

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Sanjiang Li received his B.Sc. in mathematics from Shaanxi Normal University in 1996 and his Ph.D. in mathematics from Sichuan University in 2001. He is currently a professor at the Centre for Quantum Software and Information at the University of Technology Sydney (UTS). Prior to joining UTS, he worked in the Department of Computer Science and Technology at Tsinghua University from 2001 to 2008. His primary research interests include knowledge representation, artificial intelligence, and quantum circuit compilation.